MIL-P-28582(YD)

17 March 1972

MILITARY SPECIFICATION

PRIMER COATING, EXTERIOR, LEAD PIGMENT-FREE
(UNDERCOAT FOR WOOD, READY-MIXED, WHITE AND TINTS)

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a lead pigment-free ready-mixed primer paint for previously unpainted exterior woodwork or for exterior surfaces previously painted with house paint.

1.2 Type and colors. The paint covered by this specification shall be of but one type; it may be procured, however, in white and tints (light colors), as specified by the procuring agency. (See 3.2.4 and 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking Of

TT-P-381 - Pigments-In-Oil, Tinting Color

TT-T-291 - Thinner; Paint, Volatile Spirits (Petroleum-Spirits)

TT-R-266 - Resin, Alkyd; Solutions

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

Fed. Std. No. 595 - Colors

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 8010

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

D476 - Titanium Dioxide Pigments
D2088 - Determination of Low Concentrations of Lead in Paint

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General requirements. The paint shall be ready mixed and well ground; shall not settle badly nor cake in the container; shall not change beyond minimum or maximum viscosity; and shall be readily mixed with a paddle to a uniform condition of good brushing consistency. The paint shall be free from lead pigments 1/ and other materials which will be toxic to personnel under normal conditions of use.

3.2 Qualitative requirements.

3.2.1 Condition in container. The paint shall show no excessive settling in a freshly-opened container, and shall be easily redispersed with a paddle to a smooth, homogenous state. The paint shall show no curdling, livering, caking, and shall be free from lumps or skins.

3.2.2 Dilution stability. The paint shall remain stable and uniform after reduction in the proportion of 8 parts by volume of the paint with 1 part by volume of mineral spirits conforming to TT-T-291, grade I.

3.2.3 Working properties. The paint as received shall brush easily on well-prepared wood surfaces and shall permit lapping without difficulty for a minimum of seven minutes.

3.2.4 Color. Unless otherwise specified, the paint shall be white. If a tinted paint is specified, it shall be a practical match of the specified reference standard (see 6.2).

3.2.5 Skinning. The paint shall not skin within 48 hours in a three-quarters-filled container as tested in 4.3.1.

3.2.6 Flexibility. A dried film of the paint, tested and prepared as in 4.3.2 shall withstand bending without cracking and flaking.

3.2.7 Fungus resistance. When tested as in Table III, the primer shall show no fungus growth.

3.3 Quantitative requirements. The paint shall conform to the quantitative requirements specified in Table I.

3.3.1 Pigment and vehicle requirements. The pigment and vehicle requirement shall conform to the requirement specified in Table II.

1/When tested as in ASTM D2088 the dry paint film shall contain not more than 0.5 percent lead.

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TABLE I - Quantitative Requirements of Paint

Characteristics	Requirements Minimum	Maximum
Weight, pounds per gallon of paint	10.5	-
Pigment, percent by weight of paint	42	44
Vehicle, percent by weight of paint	56	58
Pigment volume (total pigment in total nonvolatile, by volume)	39	41
Water, percent by weight of paint	-	1
Vehicle solids, percent by weight of vehicle	38	-
Gloss (85° specular)(48 hours air dry at room temperature)		50
Coarse particles and skins (residue retained on No. 325 standard sieve), percent by weight of pigment	-	2
Consistency, Krebs-Stormer, shearing rate 200 rpm:		
Grams	190	275
Equivalent K.U.	80	92
Daylight, 45°, 0° directional reflectance, white only	75	-
Dry-hard time, hours	-	18
Absorption, inch	-	1/8
Fineness of grind	3	-
Phthalic anhydride, percent by weight of paint	5	6

TABLE II - Pigment and Vehicle Requirement

Ingredients	Percent by Weight Minimum	Maximum
Pigment		
Titanium dioxide pigment (ASTM D476, Type III)[1/]	44	
Other pigments[2/]		55
Vehicle		
Flat-alkyd solids[3/]	19	
Long oil alkyd solids (TT-R-266, Type 1, Class B)	19	
Thinner, drier and mildew inhibitor		62

1/Upon analysis the extracted pigment shall show not less than the minimum value of 38 percent titanium dioxide (TiO_2).

2/Magnesium silicate, aluminum silicate, silica, or any mixture thereof. Fibrous magnesium silicate is preferred.

3/Polymerized, pure drying oil modified, having maximum phthalic anhydride content of 32 percent.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling and inspection shall be in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.3 Test procedures.

4.3.1 The tests indicated in Table III shall be conducted in accordance with Fed. Test Method Std. No. 141.

4.3.2 Flexibility. The film shall be applied to a flat tin panel with a 0.003-inch Bird film applicator (0.006-inch gap clearance) or similar doctor blade which produces the same wet-film thickness as the Bird blade, shall be air-dried for 18 hours, baked for 5 hours at 105° ± 20° C.; and bent over 1/8-inch mandrel after cooling for 1/2 hour at 23° ± 1.1° C. (73.4° ± 2°F.). See also Table III.

4.3.3 Color (tinted paints only). Draw down a film of the paint on a white opaque glass panel using an applicator that deposits a dry film thickness of approximately 1 1/2 mils. Air dry the panel at room temperature for 24-48 hours then compare the color with the reference standard in accordance with method 4250 of Fed. Test Method Std. No. 141 for compliance with 3.2.4.

TABLE III - Applicable Test Methods of Fed. Test Method Std. No. 141

Sampling and Tests	Test Method	Requirements Paragraph
Sampling	1031	4.2
Pigment, analysis of	7261	Table II
Titanium dioxide	7261, Para. 4.2.2	Table II
Other pigments	7261, Para. 4.2.4	Table II
Weight per gallon	4184	Table I
Pigment, percent by weight	4021	Table I
Pigment volume	4311	Table I
Water	4081	Table I

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TABLE III - Applicable Test Methods of Fed. Test Method Std. No. 141 (Continued)

Sampling and Tests	Test Method	Requirements Paragraph
Vehicle solids	4051	Table I
Coarse particles and skins	4091	Table I
Consistency	4281	Table I
Drying time	4061	Table I
Daylight directional reflectance	6121	Table I
Working properties	4541 1/	3.2.3
Absorption	4421	Table I
Dilution stability	4203	3.2.2
Skinning	3021	3.2.5
Fineness of grind	4411	Table I
Gloss (85° - Specular)	6103	Table I
Flexibility	2161, 6221	3.2.6
Phthalic anhydride	7021	Table I
Condition in container	3011.1	3.2.1
Color (tinted paints only)	4250	3.2.4
Fungus resistance	6271.1	3.2.7

1/Brush one-half of a panel of select grade western red cedar with the primer at a uniform spreading rate of 450 square feet per gallon. The panel should be 5 x 18 inches in size or larger. Use a 1 1/2-inch brush. After a drying time interval of 7 minutes, coat the remainder of panel. Observe working properties of the paint during the test and note whether there is difficulty in picking up the lap. (It is not necessary that the two areas blend so perfectly that no lap mark is visible).

4.4 Inspection of preparation for delivery. The packaging, packing, and marking of the primer shall be inspected to determine conformance to the requirements of Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The paint shall be packaged, packed and marked in accordance with TT-P-143. The level of packaging shall be A or C and level of packing shall be A, B, or C, as specified (see 6.2). The paint shall be furnished in 1-gallon multiple friction plug metal cans or 5-gallon lug cover steel pails, as specified (see 6.2.).

5.2 Directions for use.

5.2.1 The directions for use shall be legible, and shall be shown on the opposite side bearing the identification of the paint. (This is a priming paint for previously unpainted exterior woodwork or for exterior surfaces

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previously painted with housepaint. This priming paint should never be used as a topcoat to be left exposed to the weather for a long time. Suitable top coat should be applied within a week or two after applying this primer.)

For Previously Unpainted Wood

Recommended Procedure	Two-Coat Work	Three-Coat Work
For first coat	Apply the primer as it comes from the container, without any thinning. Use at least 1 gallon for each 450 square feet of surface.	Apply the primer. Thin if necessary with not more than 1 pint of turpentine or mineral spirits to the gallon. Use at least 1 gallon for each 600 square feet of surface.
Second coat	Use a suitable topcoat paint such as TT-P-19, MIL-P-52324 or TT-P-105. Apply it as directed on its label.	
Third coat		Apply the topcoat paint as directed on its label.

For Repainting

Recommended Procedure	One-Coat Work (old paint must be in good condition)	Two-Coat Work (old paint should be well worn)
For first coat	Use a suitable topcoat paint such as MIL-P-52324 or TT-P-105. Apply it as directed on its label. Do not use a primer.	Apply the primer. Thin, if necessary, with not more than 1 pint of thinner to a gallon. Use at least 1 gallon for each 600 square feet of surface.
Second coat		Use a suitable topcoat paint such as TT-P-19, MIL-P-52324 or TT-P-105. Apply it as directed on its label.

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6. NOTES

6.1 Intended use. This paint is intended for general exterior use for unpainted exterior woodwork or for exterior surfaces previously painted. The primer is especially useful on wood surfaces of residential housing to which children may be commonly exposed.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and level of packing required. (See 5.1)
(c) Unit quantity required. (See 5.1)
(d) Whether white or tinted paint is required. Tinted paint may be specified by reference to Fed. Std. 595 or other appropriate standard.

6.3 Basis of purchase. The paint covered by this specification description should be purchased by volume, the unit being 1 U.S. Gallon of 231 cubic inches at 15.6° C. (60°F.).

6.4 Sample paint formula. The requirements of this specification can be met by paint of the composition in TABLE IV, given both by volume and by weight.

TABLE IV - Sample Paint Formula

Ingredients	By Volume Gallons	By Weight Pounds
Titanium dioxide (ASTM D476, Type III)	.0620	2.00
Magnesium silicate (TT-P-403a)	.1080	2.50
Organophilic magnesium montmorillonite	.0020	0.03
Total pigment	.1720	4.53
Flat alkyd solution (40% solids)	.3900	2.88
Long oil alkyd solution (60% solids) (TT-R-266c, Type I, Class B)	.2500	1.92
Preservative	.0043	0.05
Liquid paint drier	.0070	0.05
Anti-skinning agent	.0010	0.01
Thinner	.1750	1.16
	.9993	10.60

Pigment volume concentration - 40%
Pigment, percent by weight of paint - 43
Vehicle, percent by weight of paint - 57

Preparing Activity
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(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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